

Part I

## Recitation Questions

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### Problem 1

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February 22, 2026

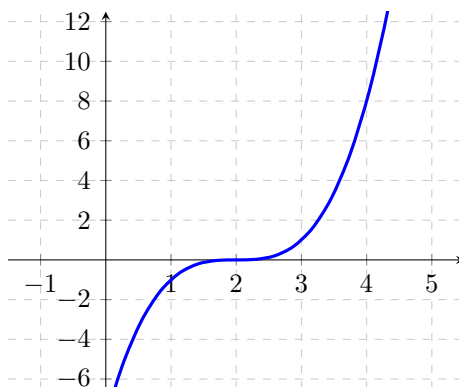
**Problem 1** Determine whether the following statements are true or false and give either an explanation or a counterexample.

- (a) The function  $f(x) = \sqrt{x}$  has a local maximum on the interval  $[0, 1]$ .

**Explanation.** False. Since  $\sqrt{x}$  is increasing over the entire region  $[0, 1]$ , the only candidate for a local maximum would be  $x = 1$ . But by definition, endpoints are never local extrema. So  $f(x) = \sqrt{x}$  has no local maximum on the interval  $[0, 1]$ .

- (b) If  $f'(2) = 0$ , then  $x = 2$  is either a local maximum or local minimum of  $f$ .

**Explanation.** False. Consider  $f(x) = (x - 2)^3$ . This has derivative  $f'(x) = 3(x - 2)^2$ , so that  $f'(2) = 0$ . We can see from the graph of  $f$  below, that  $x = 2$  is not a local extremum for  $f$ .



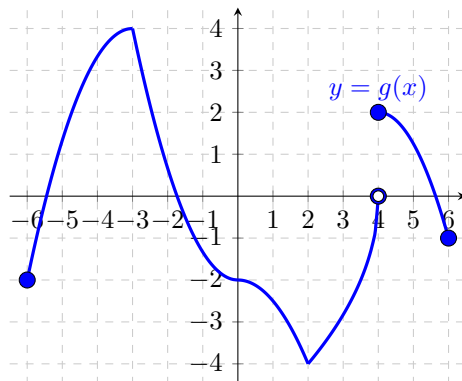
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## Problem 2

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**Problem 2** The entire graph of a function  $g$  is given below.



Based on the graph of  $g$ , answer the questions below.

- (a) List the  $x$ -coordinates of all critical points of  $g$ .  
**Explanation.**  $x = -3$ ,  $x = 0$ ,  $x = 2$ , and  $x = 4$ .
- (b) List the  $x$ -coordinates of all critical points of  $g$  where  $g'(x) = 0$ .  
**Explanation.**  $x = 0$
- (c) List the  $x$ -coordinates of all critical points of  $g$  where  $g'(x)$  is **undefined**.  
**Explanation.**  $x = -3$ ,  $x = 2$ , and  $x = 4$ .
- (d) List the  $x$ -coordinates of all local maximums of  $g$ .  
**Explanation.**  $x = -3$  and  $x = 4$ .
- (e) List the  $x$ -coordinates of all local minimums of  $g$ .  
**Explanation.**  $x = 2$
- (f) List all intervals where  $g$  is both decreasing AND concave down.  
**Explanation.**  $(0, 2)$  and  $(4, 6)$ .
- (g) List all intervals where  $g$  is both decreasing AND concave up.  
**Explanation.**  $(-3, 0)$
- (h) List all intervals where  $g$  is both increasing AND concave down.  
**Explanation.**  $(-6, -3)$
- (i) List all intervals where  $g$  is both increasing AND concave up.  
**Explanation.**  $(2, 4)$
- (j) List the  $x$ -coordinates of all inflection points of  $g$ .  
**Explanation.**  $x = -3$ ,  $x = 0$ , and  $x = 2$ .

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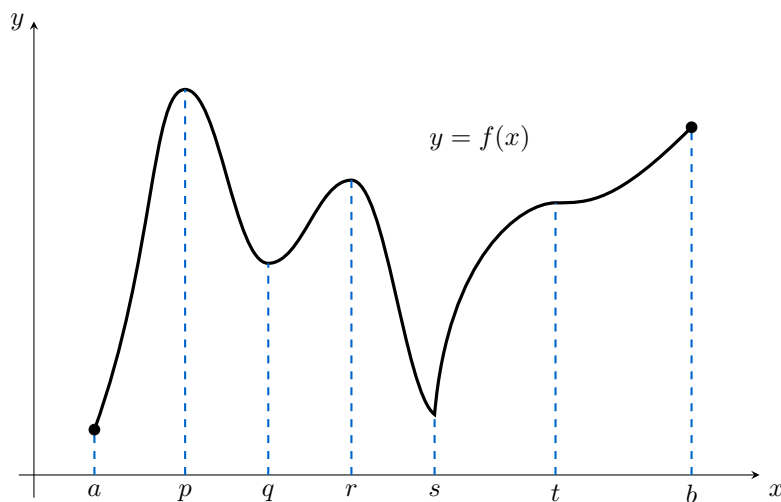
# Problem 3

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February 22, 2026

Problem 3

**Problem 3** For each point in the interval  $(a,b)$  and identified on the graph below, determine if the function  $f$  has a critical point, a local max or min at that point.



**Explanation.** (p) The function  $f$  has a critical point and a local maximum at  $x = p$ .

(q) The function  $f$  has a critical point and a local minimum at  $x = q$ .

(r) The function  $f$  has a critical point and a local maximum at  $x = r$ .

(s) The derivative of  $f$  does not exist at  $x = s$  because the function  $f$  has a corner at  $x = s$ . The function has a critical point and a local minimum at  $x = s$ .

(t) The function  $f$  has a critical point but not a local maximum or minimum at  $x = t$ .

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# Problem 4

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**Problem 4** Locate the critical points and use the second derivative test to determine whether they correspond to local maxima or local minima. **EXPLAIN.**

$f(x) = (x + c)^4$  where  $c$  is a positive constant

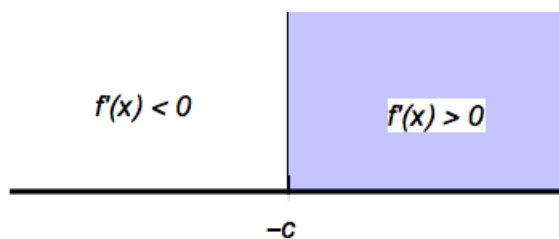
**Explanation.**

$$f'(x) = 4(x + c)^3, f''(x) = 12(x + c)^2$$

To find the critical points:  $f'(x) = 0 = 4(x + c)^3$ . This occurs when  $x = -c$

Using the second derivative test:  $f''(-c) = 12(-c + c)^2 = 0$ . The second derivative test was inconclusive so we need to use the first derivative test.

Making a sign chart we see that if we plug something a little less than  $-c$  into  $f'$ , we'll get a negative number. If we plug something a little greater than  $-c$  into  $f'$ , we'll get a positive number.



The derivative of  $f$ ,  $f'$ , does not change sign on the intervals  $(-\infty, -c)$  and on  $(-c, \infty)$ , therefore, we have the above chart. From the chart, we can conclude that  $f$  has a local minimum at  $x = -c$ , and that  $f$  also has a global minimum there. We could also have used our understanding of the family of functions  $g(x) = x^4$  to reason  $x = -c$  would be a minimum.

At  $x = -c$ ,  $f$  has a local minimum by the first derivative test since  $x = -c$  is a critical point of  $f$  and the sign of  $f'$  changes from negative to positive around  $x = -c$ .



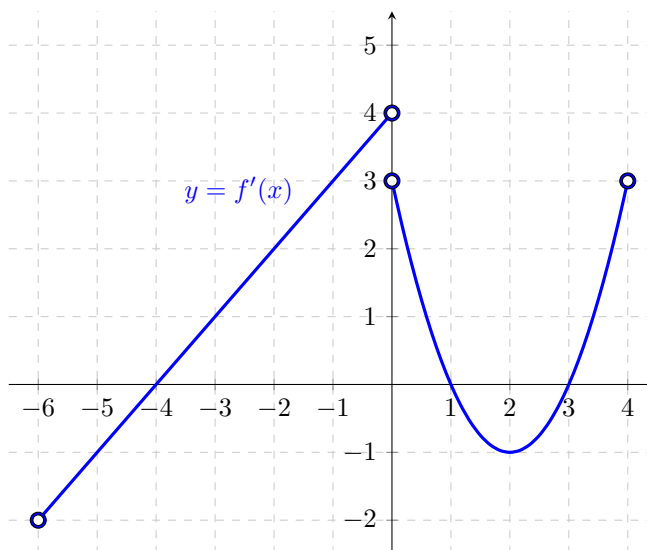
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# Problem 5

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February 22, 2026

**Problem 5** Let  $f$  be a function that is continuous on its domain  $(-6, 4)$ . The graph of  $f'$ , the derivative of  $f$ , is given below.



Based on the graph of  $f'$ , answer the question below.

- (a) List x-coordinates of all critical points of  $f$ .

**Explanation.** Notice,  $f'(-4) = f'(1) = f'(3) = 0$ , and  $f'(0)$  is not defined. Therefore, the function  $f$  has critical points at  $x = -4$ ,  $x = 0$ ,  $x = 1$  and  $x = 3$ .

- (b) List x-coordinates of all critical points of  $f$  where  $f'(x) = 0$ .

**Explanation.**  $x = -4$ ,  $x = 1$  and  $x = 3$ .

- (c) List x-coordinates of all critical points of  $f$  where  $f'(x)$  is **undefined**.

**Explanation.**  $x = 0$

- (d) List x-coordinates of all local minimums of  $f$ .

**Explanation.**  $x = -4$  and  $x = 3$ . At these points the sign of  $f'(x)$  changes from negative to positive.

- (e) List x-coordinates of all local maximums of  $f$ .

**Explanation.**  $x = 1$  At this point the sign of  $f'(x)$  changes from positive to negative.

Problem 5

- (f) List all intervals where  $f$  is decreasing and concave down.

**Explanation.**  $(1, 2)$  Notice,  $f'(x)$  is negative and decreasing there.

- (g) List all intervals where  $f$  is decreasing and concave up.

**Explanation.**  $(-6, -4)$  and  $(2, 3)$ . Notice,  $f'(x)$  is negative and increasing there.

- (h) List all intervals where  $f$  is increasing and concave down.

**Explanation.**  $(0, 1)$  Notice,  $f'(x)$  is positive and decreasing there.

- (i) List all intervals where  $f$  is increasing and concave up.

**Explanation.**  $(-4, 0)$  and  $(3, 4)$ . Notice,  $f'(x)$  is positive and increasing there.

- (j) List x-coordinates of all inflection points of  $f$ .

**Explanation.**  $x = 0$ , and  $x = 2$   $f'(x)$  is increasing on  $(-6, 0)$  and decreasing on  $(0, 2)$ , then increasing on  $(2, 4)$ .

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# Problem 6

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February 22, 2026

**Problem 6** Find the critical points of  $f$  on the given interval. Determine whether the function  $f$  has a local maximum, local minimum or no local extremum at each critical point. **EXPLAIN.**

(a)  $f(x) = x\sqrt{2-x^2}$  on  $(-\sqrt{2}, \sqrt{2})$ .

**Explanation.**

$$\begin{aligned} f'(x) &= \sqrt{2-x^2} + x \left( \frac{1}{2\sqrt{2-x^2}}(-2x) \right) \\ &= \sqrt{2-x^2} - \frac{x^2}{\sqrt{2-x^2}} \\ &= \frac{2-x^2-x^2}{\sqrt{2-x^2}} \\ &= \frac{2(1-x^2)}{\sqrt{2-x^2}} \end{aligned}$$

Critical points of  $f$  occur where  $f'(x) = 0$  or where  $f'(x)$  does not exist. Solving  $f'(x) = 0$  yields that  $2(1-x^2) = 0$ , or  $x = \pm 1$ .  $f'(x)$  does not exist when  $2-x^2 \leq 0$ , but the points  $x = \pm\sqrt{2}$  are not in the domain.

Let us use the Second Derivative Test to determine whether the function  $f$  has a local maximum or local minimum at the points where  $x = \pm 1$ .

$$f''(x) = \frac{d}{dx} \frac{2(1-x^2)}{\sqrt{2-x^2}} = \frac{2x(x^2-3)}{(2-x^2)^{\frac{3}{2}}}$$

Since  $f''(-1) = 4$ , the point  $(-1, -1)$  is a local minimum by the Second Derivative Test. Since  $f''(1) = -4$ , the point  $(1, 1)$  is a local maximum by the Second Derivative Test.

(b)  $f(x) = x^3e^{-x}$  on  $(-1, 5)$ .

**Explanation.**

$$\begin{aligned} f'(x) &= 3x^2e^{-x} + x^3(-e^{-x}) \\ &= x^2e^{-x}(3-x) \end{aligned}$$

Notice that  $f'(x)$  always exists, and so all of the critical points of  $f$  occur when  $f'(x) = 0$ . Solving this equation:

$$\begin{aligned} x^2e^{-x}(3-x) &= 0 \\ x^2(3-x) &= 0 \\ x = 0 \quad \text{or} \quad x &= 3 \end{aligned}$$

It is easier to use the First derivative test in order to determine whether the function  $f$  has a local maximum, local minimum or no local extremum at the point where  $x = 0$ . Since  $-1 < 0 < 1$ ,  $f'(-1) = 4e > 0$ , and  $f'(1) = 2e^{-1} > 0$ , it follows that the sign of  $f'$  does not change at  $x = 0$ . The function  $f$  has **no local extremum** at  $x = 0$ .

Similarly, we will use the First derivative test in order to determine whether the function  $f$  has a local maximum, local minimum or no local extremum at the points where  $x = 3$ . Since  $1 < 3 < 4$ ,  $f'(1) = 2e^{-1} > 0$ , and  $f'(4) = -16e^{-4} < 0$  it follows that the sign of  $f'$  changes from positive to negative at  $x = 3$ . The function  $f$  has **a local maximum** at  $x = 3$  by the First Derivative Test.

(c)  $f(x) = x \ln \left( \frac{x}{5} \right)$  on  $(0, 5)$ .

**Explanation.**

$$\begin{aligned} f'(x) &= \ln \left( \frac{x}{5} \right) + x \cdot \frac{5}{x} \cdot \frac{1}{5} \\ &= \ln \left( \frac{x}{5} \right) + 1 \end{aligned}$$

Notice that  $f'(x)$  exists for all values in  $(0, 5)$ , and so all of the critical points of  $f$  occur when  $f'(x) = 0$ . Solving this equation:

$$\begin{aligned} \ln \left( \frac{x}{5} \right) + 1 &= 0 \\ \ln \left( \frac{x}{5} \right) &= -1 \\ \frac{x}{5} &= e^{-1} \\ x &= 5e^{-1} = \frac{5}{e} \end{aligned}$$

Let's use the Second Derivative Test to determine whether the function  $f$  has a local maximum, local minimum or no local extremum at the points where  $x = \frac{5}{e}$ .

$$\begin{aligned} f''(x) &= \frac{d}{dx} \left( \ln \left( \frac{x}{5} \right) + 1 \right) \\ &= \frac{d}{dx} (\ln(x) - \ln(5) + 1) = \frac{1}{x} \end{aligned}$$

Since,  $f''\left(\frac{5}{e}\right) > 0$ , the point  $\left(\frac{5}{e}, -\frac{5}{e}\right)$  is a local minimum by the Second Derivative Test.

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# Problem 7

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February 22, 2026

**Problem 7** Let  $f(x) = \frac{1}{1+x^2}$ . Find the following for  $f$ :

- (a)  $f'$  and  $f''$

**Explanation.**

$$\begin{aligned} f'(x) &= \frac{(1+x^2)(0) - 1(2x)}{(1+x^2)^2} \\ &= \frac{-2x}{(1+x^2)^2} \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{(1+x^2)^2(-2) - (-2x)(2)(1+x^2)(2x)}{(1+x^2)^4} \\ &= \frac{-2(1+x^2) + 8x^2}{(1+x^2)^3} \\ &= \frac{6x^2 - 2}{(1+x^2)^3} \end{aligned}$$

- (b) *Critical points*

**Explanation.** Since  $1+x^2 > 0$  for all  $x$ ,  $f$  is differentiable over all real numbers. Thus all critical points of  $f$  occur when  $f'(x) = 0$ . But a fraction equals 0 if and only if its numerator equals 0. So

$$f'(x) = 0 \quad \implies \quad -2x = 0 \quad \implies \quad x = 0$$

Hence, the only critical point is  $x = 0$ .

- (c) *Local extrema (and check your answers with both the first and second derivative tests)*

**Explanation.** At  $x = 0$ ,  $f'$  changes sign from positive to negative. Thus  $f$  goes from increasing to decreasing, and therefore by the first derivative test  $x = 0$  is a local maximum of  $f$ .

For the second derivative test, we have that

$$f''(0) = \frac{6(0)^2 - 2}{(1+0^2)^3} = \frac{-2}{1} = -2 < 0$$

and thus we again conclude that  $x = 0$  is a local maximum of  $f$ .

- (d) *Inflection points. EXPLAIN.*



**Explanation.** By the results in part

$$\begin{aligned} f''(x) &= \frac{6x^2 - 2}{(1 + x^2)^3} \\ &= \frac{6(x^2 - \frac{1}{3})}{(1 + x^2)^3} \\ &= \frac{6(x - \frac{1}{\sqrt{3}})(x + \frac{1}{\sqrt{3}})}{(1 + x^2)^3} \end{aligned}$$

which implies that the only candidates for inflection points are points where  $x = \pm \frac{1}{\sqrt{3}}$ . On the other hand, since  $-1 < -\frac{1}{\sqrt{3}} < 0$ ,  $f''(-1) = \frac{1}{2}$ , and  $f''(0) = -2$ , it follows that  $f''$  changes the sign at  $x = -\frac{1}{\sqrt{3}}$ . Similarly, since  $0 < x = \frac{1}{\sqrt{3}} < 1$ ,  $f''(0) = -2$ ,  $f''(1) = \frac{1}{2}$ , it follows that  $f''$  changes the sign at  $x = \frac{1}{\sqrt{3}}$ , too. Since,

$$\begin{aligned} f\left(\frac{1}{\sqrt{3}}\right) &= \frac{3}{4} \\ f\left(-\frac{1}{\sqrt{3}}\right) &= \frac{3}{4} \end{aligned}$$

The function  $f$  has inflection points at  $\left(-\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$ ,  $\left(\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$  because  $f$  is continuous at those points and  $f$  changes concavity around those points.

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# Problem 8

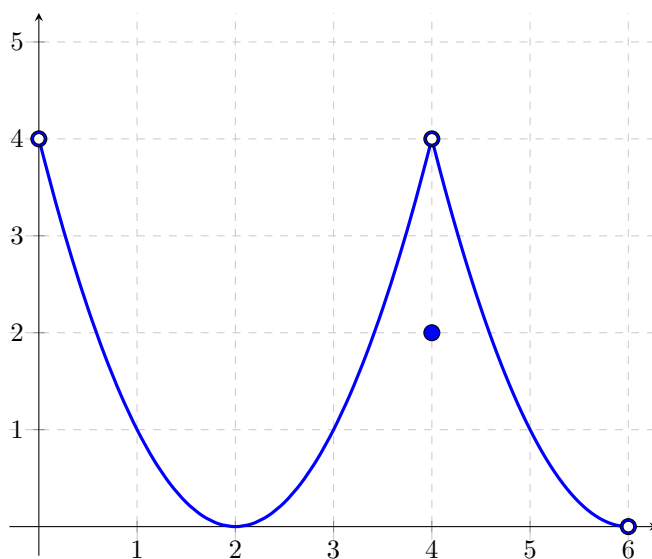
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February 22, 2026

**Problem 8** Sketch a possible graph of a function  $f$  that has the following properties:

- (a)  $f$  is defined on the interval  $(0, 6)$ .
- (b)  $f$  has no local maximums.
- (c)  $f$  has exactly two local minimums.

**Explanation.** This could be the graph of  $f$ . Notice two local minimums: at  $x = 2$  and at  $x = 4$ .



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# Problem 9

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February 22, 2026

**Problem 9** Consider the parabola  $f(x) = ax^2 + bx + c$  where  $a, b, c$  are constants. For what values of  $a, b, c$  is  $f$  concave up? For what values of  $a, b, c$  is  $f$  concave down?

**Explanation.**

$$\begin{aligned}f'(x) &= 2ax + b \\f''(x) &= 2a\end{aligned}$$

This means the sign of  $a$  will determine whether  $f$  is concave up or down. When  $a < 0$ ,  $f$  is concave down, which makes sense because then the graph of  $f$  is a downward opening parabola. When  $a > 0$ ,  $f$  is concave up, which makes sense because then the graph of  $f$  is an upward opening parabola. If  $a = 0$ ,  $f$  has no concavity because it is a linear function.